
--- MATH135A Project 2: Problem 1 ---

Solves a 4x4 system using Gaussian elimination with scaled partial pivoting

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% --- 1. Setup ---
A = [ 0.4096, 0.1234, 0.3678, 0.2943;
      0.2246, 0.3872, 0.4015, 0.1129;
      0.3645, 0.1920, 0.3781, 0.0643;
      0.1784, 0.4002, 0.2786, 0.3927];

B = [ 0.4042; % Using the value 0.4042 as you provided
      0.1250;
      0.4255;
      0.2557];

% Get the size of the matrix
n = 4;

% Create the augmented matrix
Aug = [A, B];

% Create the scale vector (as a column)
% We can also calculate this in MATLAB:
s = max(abs(A), [], 2); % Finds max absolute value in each row

disp('Initial Augmented Matrix:');
disp(Aug);
disp('Initial Scale Vector:');
disp(s);

% --- 2. Forward Elimination ---
for k = 1:n-1

    % --- Pivoting ---
    % Find scaled ratios for rows k through n in column k
    ratios = abs(Aug(k:n, k)) ./ s(k:n);

    % Find the index of the largest ratio
    [max_val, max_idx] = max(ratios);

    % Calculate the actual row number to pivot with
    pivot_row = k + max_idx - 1;

    % Swap rows if pivot_row is different from the current row k
    if pivot_row ~= k
        % Swap rows in Aug matrix
        temp_row = Aug(k, :);
        Aug(k, :) = Aug(pivot_row, :);
        Aug(pivot_row, :) = temp_row;

        % IMPORTANT: Swap elements in the scale vector too
        temp_s = s(k);
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        s(k) = s(pivot_row);
        s(pivot_row) = temp_s;
    end

    % --- Elimination ---
    % Loop through all rows below the pivot row k
    for i = k+1:n

        % Find the multiplier
        m = Aug(i, k) / Aug(k, k);

        % Subtract the scaled pivot row from the current row i
        Aug(i, :) = Aug(i, :) - m * Aug(k, :);
    end
end

disp('Upper Triangular (Eliminated) Matrix:');
disp(Aug);

% --- 3. Back Substitution ---
% Create an empty solution vector (filled with zeros)
x = zeros(n, 1);

% Solve for the last element, x(n)
x(n) = Aug(n, n+1) / Aug(n, n);

% Loop backwards from n-1 down to 1
for k = n-1:-1:1

    % Calculate the sum of the parts we already know
    % (Aug(k, k+1:n) is a row vector, x(k+1:n) is a column vector)
    sum_of_known_parts = Aug(k, k+1:n) * x(k+1:n);

    % Solve for x(k)
    x(k) = (Aug(k, n+1) - sum_of_known_parts) / Aug(k, k);
end

% --- 4. Final Answer ---
disp('Final Answer Vector x:');
disp(x);

Initial Augmented Matrix:
    0.4096    0.1234    0.3678    0.2943    0.4042
    0.2246    0.3872    0.4015    0.1129    0.1250
    0.3645    0.1920    0.3781    0.0643    0.4255
    0.1784    0.4002    0.2786    0.3927    0.2557

Initial Scale Vector:
    0.4096
    0.4015
    0.3781
    0.4002

```

Upper Triangular (Eliminated) Matrix:

0.4096	0.1234	0.3678	0.2943	0.4042
0	0.3465	0.1184	0.2645	0.0797
-0.0000	0	0.0906	-0.2924	-0.1701
0.0000	0	0	-0.1871	0.0895

Final Answer Vector x:

3.8718
1.7649
-3.4220
-0.4787

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